



NATIONAL JUNIOR COLLEGE
2009 SH2 PRELIMINARY EXAMINATIONS

MATHEMATICS
Higher 1

8863/01

Additional Materials: Answer Paper
List of Formulae (MF15)
Cover Sheet

3 hours

16 September 2009, Wednesday

0815 – 1115 hrs

INSTRUCTIONS TO CANDIDATES

Write your name, registration number, subject tutorial group, on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for diagrams or graphs.

Do not use paper clips, highlighters, glue or correction fluid.

Answer **ALL** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in the brackets [] at the end of each question or part question.

At the end of the examination, fasten all your work securely together.

This question paper consists of 8 printed pages

[Turn over

Section A: Pure Mathematics [40 Marks]

- 1 The curve C has the equation $y = \ln\left(\frac{1}{x}\right)$. Find the equation of the tangent at the point $P(2, -3)$. [2]
- 2 The giant water trough shown in Figure 1 below has a length of 8 m and a height of $\sqrt{3}$ m. Its ends are in the shape of equilateral triangles with $AB = BC = AC = 2$ m. A cross-section of the trough (with some water) is shown in Figure 2.

The trough, initially empty, is to be filled with water from a pipe through vertex **A** which delivers water at a constant rate of $0.3 \text{ m}^3 \text{ s}^{-1}$. The height of water in the trough is denoted by h m and the width of the water surface (DE) is denoted by x m.

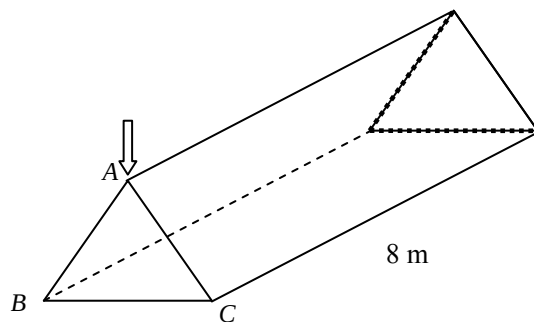


Figure 1

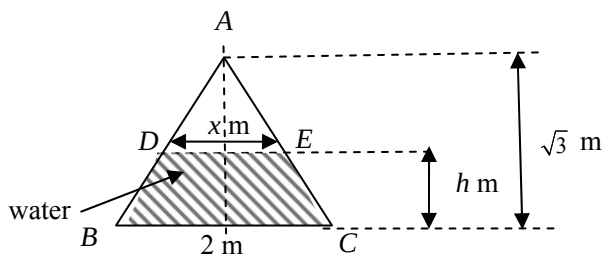


Figure 2

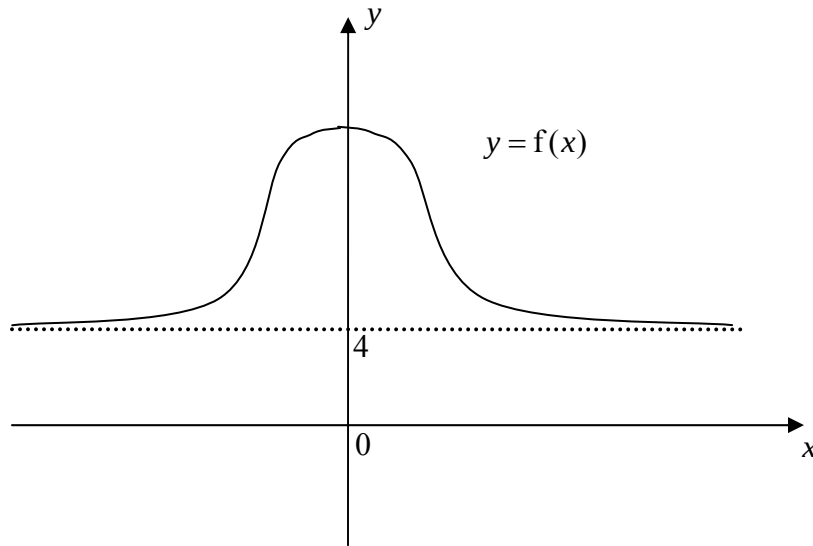
- (i) Show that $x = \frac{2}{\sqrt{3}}(\sqrt{3} - h)$. [1]
- (ii) Hence, find the rate at which the height is increasing when the height of the water is 0.5 m. [3]

- 3(a)** It is given that $g(x) = \frac{3x+1}{x+2}$, where a and b are constants.

State the equations of the asymptotes.

[1]

(b)



The diagram above shows a sketch of the curve $y = f(x)$, where $y = 4$ is an asymptote and the maximum value of y occurs at $x = 0$.

Sketch the graph of $y = f'(x)$.

[2]

- (c)** A continuous curve $y = f(x)$, $x \in \mathbb{R}$ satisfies the following conditions:

- i)** f is strictly decreasing for $-2 < x < 3$,
- ii)** f has exactly two stationary points at $(-2, 5)$ and $(3, 1)$.

Sketch the curve $y = f(x)$.

[2]

- 4**
- (i)** Sketch the graph of $(x-3)^2 + (y-1)^2 = 1$, stating clearly any axial intercepts. [2]
 - (ii)** State the line(s) of symmetry for the above graph. [1]
 - (iii)** Find the range of values of k for which the graph and the line $y = kx + 1$ intersect at 2 points. [2]

5 Sketch, on a single diagram, the graph of

$$f(x) = \cos \frac{x}{2} \text{ for } 0 \leq x \leq \pi, \text{ and } g(x) = 2 \cos^{-1}(x) \text{ for } 0 \leq x \leq 1,$$

stating clearly any intersection(s) with the axes. [3]

(i) Write down the period of $f(x)$. [1]

(ii) State, algebraically, the relationship between the graph of f and g . [1]

(iii) State the set of values of x for which $f(x) \leq g(x)$. [1]

6 The functions f , g and h are defined for $x \in \mathbb{R}$ by

$$f : x \mapsto |x^2 - 4|, \quad x > 0,$$

$$g : x \mapsto 3 \ln(x+1), \quad x > -1,$$

$$h : x \mapsto 7 + \frac{x^2}{2}, \quad x > 0.$$

(i) Show that the composite function $gh(x)$ exists and state its range. [3]

(ii) Explain why f has no inverse function. State a restricted domain of f in the form of $[a, \infty)$ for f^{-1} to exist. [2]

(iii) Using your restricted domain in (ii), define, in a similar form, the function f^{-1} . [2]

(iv) Sketch the graph of ff^{-1} . [2]

7(a) Find

(i) $\int \frac{(e^x + e^{-x})^2}{e^{2x}} dx,$ [2]

(ii) $\int \sqrt{(3x+2)^4} dx.$ [2]

(b) (i) Find the gradient function of the curve $y = 4 - (x-1)^2$. [2]

(ii) Hence, find the area bounded by the gradient function of the curve and the curve itself. [3]

Section B: Statistics [60 Marks]

- 8** In order to find out the percentage of students who will still travel overseas despite the H1N1 outbreak, a group of four undergraduates decides to sample 100 students. During allocation of task, the four undergraduates A , B , C and D decided that each of them will sample 25 students, and taking their samples from a junior college, a polytechnic, a secondary and a primary school respectively.

- (i) Name the sampling model chosen by these four undergraduates. [1]
- (ii) Identify a significant area of weakness in their above proposed model and suggest improvements to the model which will address the weakness. [3]

- 9** Seven balls of which three are white and four are black are in a box. Three balls are randomly picked and put in bag A .

Find the probability that there are

- (i) exactly three white balls in bag A , [2]
- (ii) all black balls in bag A , [2]
- (iii) exactly three white balls in bag A , given that at least one white ball has been drawn and put in bag A . [3]

Suppose it is now known that bag A has exactly two white balls and one black ball and that all the remaining balls are then put in a bag B .

- (iv) A bag is selected at random and one ball is drawn from it. Find the probability that the ball is white. [2]

- 10** In year 2008, a survey was done by the Ministry of Manpower (MOM) to determine the amount of daily wages, denoted as X , that part-time workers were receiving. Past records show that on average, the wages that each worker received was \$30. Data from 90 random workers are collected and their wages are summarized as shown below:

$$\sum x = 2790, \quad \sum (x - 31)^2 = 1947.$$

- (i) Calculate the unbiased estimate of the mean and variance of X . [2]
- (ii) An audit company suspects that a particular restaurant is underestimating the mean wage of its part-time workers per day. Conduct an appropriate test at 5% level of significance. [4]
- (iii) State, giving a reason, whether it is necessary for the wages to follow a normal distribution for the test to be valid. [1]

A month later, MOM wants to find out if there is a change in the mean wage and decides to do a test on the collected data as above. Find the range of values of α if MOM were to conclude that there is a change in the mean wage at α % level of significance. [3]

- 11** In Brook University, there are 150 professors. On average, 2 out of every 10 professors are diagnosed with flu weekly.

- (i) Find the probability that less than 28 professors are diagnosed with flu in a week. [2]
- (ii) Find the least possible value of n such that the probability of at most n professors being diagnosed with flu weekly exceeds 0.6. [3]
- (iii) Using a suitable approximation, find the probability that at most 37 professors are diagnosed with flu in a week. [3]
- (iv) Find the probability that the mean number of professors who are diagnosed with flu weekly is between 25.5 and 32(inclusive) over a period of 50 weeks? [3]

- 12** At a clinical laboratory, a machine is used to measure the growth of a certain bacteria at fixed time intervals and the results are tabulated as follow:

Time (days), x	5	10	24	35	48	55	67
Bacteria Growth (thousands), y	1.3	42.0	14.8	30.1	60.8	81.3	98.6

It was discovered that one of the results may be wrong. By drawing a scatter diagram, identify the result that is most likely to be incorrect. Justify your answer. [3]

The data with the correct result yield the following regression lines:

$$y + 14.16830 = 1.61302x \quad \text{and} \quad x - 0.59168y = 9.97265$$

- (i) Identify, giving a reason, the regression line of y on x and the regression line of x on y . [3]
- (ii) Find the value of the correct result, correct to 1 decimal place. [3]
- (iii) State, giving a reason, which of the least squares regression lines, y on x or x on y , should be used to express a possible linear relation between y and x . [2]
- (iv) Using the regression line you have chosen in (ii), estimate the bacteria growth in 50 days, giving your answer to the nearest thousands. Comment on its reliability. [2]

- 13** On Crabby Island, a fishmonger rears many types of live seafood in his kelong, one of which is a particular type of crab named the Hairy Crab. The weight H , in kg, of a random Hairy Crab follows a normal distribution with mean μ kg and standard deviation σ kg.

Given that $P(H < 1.5) = P(H > 3.5) = 0.33$, state the value of μ . [1]

- (i) Show that $\sigma = 2.27$. [2]
- (ii) Find the probability that out of three randomly chosen crabs, two weigh between 1.5 kg and 3.5kg each and one weighs more than 3.5kg. [3]
- (iii) Find the probability that out of 100 Hairy Crabs, at least 80 will weigh more than 1.5 kg. [3]

Another type of crab named Flower Crab found on Crabby Island follows an independent normal distribution with mean weight 3.2 kg and standard deviation 1.3 kg. Find the largest value of a such that the probability of the average weight of two randomly chosen Hairy Crabs and three randomly chosen Flower Crabs exceeds a kg is more than 0.6. [4]

End of Paper



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Write your name, registration number, subject tutorial group and subject tutor's name in the spaces provided on the cover sheet and attach it on top of your answer paper.

Circle the questions you have attempted and arrange your answers in **NUMERICAL ORDER**.

Name: _____

Registration Number: _____

Subject Tutorial Group: _____

Subject Tutor: _____

Section A	Marks	Max. Marks	Section B	Marks	Max. Marks
1		2	8		4
2		4	9		9
3		5	10		10
4		5	11		11
5		6	12		13
6		9	13		13
7		9	Subtotal		60
Subtotal		40	Presentation	-1 / -2	
Total		100	Grade		